

# Final Report: Supervised Machine Learning Classification Analysis

## 1. Data Description

The dataset used for this analysis is the **Iris dataset**, which contains 150 samples of iris flowers from three species: *Setosa*, *Versicolor*, and *Virginica*. Each sample includes four features:

- sepal length (cm)
- sepal width (cm)
- petal length (cm)
- petal width (cm)

These features were used as input variables to train and evaluate various classification models. The target variable is the species of the iris flower. This dataset is commonly used to demonstrate supervised machine learning classification tasks.

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## 2. Objective of Analysis

The main objective of this analysis is to **classify iris flowers by species** using supervised machine learning models and identify which model provides the best predictive performance. The goal is to gain insights into feature importance and understand how different models can be applied to solve classification problems.

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## 3. Model Selection and Comparison

Three supervised classification models were evaluated:

1. **Decision Tree Classifier** – A simple model that splits the feature space into axis-aligned regions.
2. **Random Forest Classifier** – An ensemble of decision trees that reduces overfitting and improves generalization.
3. **Logistic Regression** – A parametric model that estimates probabilities for each class using linear decision boundaries.

**Comparison Metrics** included:

- Accuracy

- Precision
- Recall
- F1-score

The **Random Forest Classifier** achieved the highest accuracy on the test set, indicating strong generalization while reducing variance compared to a single decision tree.

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## 4. Key Findings

- The most important features for classification were **petal length** and **petal width**, which separated the three species effectively.
  - Decision Trees were highly interpretable but prone to overfitting on the training data.
  - Random Forest improved performance by aggregating multiple trees and introducing feature randomness.
  - Logistic Regression performed reasonably well but struggled with non-linear boundaries between species.
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## 5. Limitations and Next Steps

- **Limitations:**
    - Small dataset size may limit generalization.
    - Decision Trees may overfit without proper regularization.
    - Logistic Regression may not capture complex patterns in the data.
  - **Next Steps:**
    - Collect more data to improve model robustness.
    - Explore hyperparameter tuning for Random Forest (e.g., number of trees, max depth).
    - Consider boosting methods (e.g., Gradient Boosting) for further performance improvement.
    - Implement cross-validation to validate model performance on unseen data.
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## 6. Evaluation Metrics Purpose

Evaluation metrics such as accuracy, precision, recall, and F1-score are used to **assess model performance** on both training and test datasets. They provide insight into how well the model predicts the correct class and highlight trade-offs between sensitivity and specificity.

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## 7. Observations on Supervised ML Classification

- Classification models map input features to discrete output classes.
  - Overfitting occurs when a model performs well on training data but poorly on test data.
  - Ensemble models like Random Forest reduce overfitting by combining multiple learners.
  - Proper evaluation and hyperparameter tuning are essential for selecting the best model.
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## 8. Considerations When Comparing Models

When comparing classification models, consider:

- Model accuracy on test data
  - Robustness and overfitting risk
  - Interpretability and complexity
  - Training time and computational resources
  - Feature importance and insights provided
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## 9. Common Issue: Overfitting

If a supervised classification model performs well on training data but poorly on test data, the most likely issue is **overfitting**, where the model captures noise in the training data instead of general patterns.